GHANASHYAM MANDAL | CH23B015 | INDIAN INSTITUTE OF TECHNOLOGY MADRAS

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
B. Tech in Chemical Engineering	Indian Institute of Technology, Madras	8.19	2027
Class XII (CBSE[1])	Jawahar Navodaya Vidyalaya, Bundi	90.2 %	2023
Class X (CBSE[1])	Jawahar Navodaya Vidyalaya, Koraput	93.8 %	2021

SCHOLASTIC ACHIEVEMENTS

- **Qualified JNVST**, securing merit-based selection in a highly competitive national exam for fully funded residential schooling at JNV.
- **Qualified** for the highly selective **Dakshana Foundation's JEE[2] Scholars Program** in 2021, and secured **one** of **~300-350** nationwide seats from **20,000+** of applicants through the rigorous **NDST[3] (<1% selection rate)** , which is conducted by the Dakshana Foundation.

INTERNSHIPS AND WORK EXPERIENCE

ANALYTICAL INTERN L'OREAL (May' 26 - July'26)

Replaced subjective manual testing with a data-driven scoring model — improving speed and consistency of decisions.

- **Built** a **scoring model** linking the physical measurements to consumer-perceived quality, enabling objective and reliable decisions.
- Stress-tested **6** formulations across extreme conditions (5 to 55 degrees), mitigating possible risks of field failures after deployment
- Proposed an early-warning framework to **predict** instability, reducing dependency on the quality checks & making up with trends.

PROCESS RESEARCH INTERN (July'26 - Present)

Diagnosing the core bottleneck in an industrial absorption process to unlock efficiency gains with direct cost impact.

- Identified the key rate-limiting step constraining the process performance, and redirecting effort towards the highest-leverage fix.
- Evaluated a new emulsion-based technical approach against the cost, stability, and the feasibility to help guiding a go/no-go call.
- Evaluating formulation **trade-offs** against process constraints, translating the **lab-scale gains** into **cost** and pumping implications.

COURSES AND SKILLS

Process and Product Design*	Sustainable Plastics	Machine Learning for Molecular Stimulations
Principles of Economics	Business Analytics	Google Data Analytics Certification
Programming in Python	**Database Management:** MySQL,BigQuery	Data analytics and Visualisation
Programming Languages: Python, C, R	**Visualisation Tools :** Tableau , Rstudio, PowerBI	**AI Tools:** ChatGPT, Gemini, Perplexity, Claude

ANALYTICAL AND STRATEGY PROJECTS

EV Charging ME case (India)

Market-entry analysis for an EV charging network in India, sizing demand and identifying priority states for strategic expansion.

- Quantified **43K+** public charging points needed, by sizing out the full **TAM/SAM/SOM** for an EV charging network's entry into India.
- **217%** YoY growth in 4-wheeler EV registration uncovered via analysis of **2M+** vehicle records,reshaping the charging demand thesis
- Karnataka emerged as the top pilot state after ranking all states by EV volume and penetration to guide the market-entry strategy.

Mobility Analytics for Retail & Marketing Strategy

Mobility data analysis uncovers key behavioral trends, informing strategic marketing & business, adeptly boosting engagement.

- Analyzed mobility data to uncover the key behavioral **trends** with direct implications for marketing **strategy** and business **growth.**
- Applied statistical modeling and cluster analysis to map movement patterns,capturing **82%** of movement variance across segments
- Identified **91%** of high-mobility zones overlapped with retail hotspots, informing targeted marketing & store placement strategies.

GESTURAL MUSIC (Envisage[4])

Developed an advanced finger detection model using OpenCV and mediapipe, adeptly Capable of converting gestures to music.

- Engineered an interface by integrating **Pygame MIDI** with custom mappings for conversion of movements into a musical output.
- Utilized **OpenCV** for real-time video capture & processing, and enabled dynamic computer vision interactions and visual feedback.
- Designed a custom gesture-to-**MIDI mapping system**, and adeptly translated dynamic finger positions into discrete musical notes

Molecular Optimization (CH5003)

Proposed** an AI-driven pipeline to optimize a **single drug** candidate for **dual-target** disease treatment under **Dr. Abhinav S. Raman

- Proposed a multi-objective scoring model to rank competing options against the constraints like safety, cost, & manufacturability.
- Structured a complete evaluation pathway from rigorous broad initial screening down to one prioritized final recommendation.
- Balanced technical feasibility against real-world manufacturing constraints to identify the most viable and efficient option overall.

POSITION OF RESPONSIBILITY

E-CELL[5]

MANAGER M&PR[6] (May' 24- Mar' 25)

Co-managed and executed initiatives, enhancing brand reputation & visibility via targeted campaigns & media engagement.

- Contributed to the strategic planning and seamless execution of **3+** major events, coordinating five+ teams for **2,000+** audience.
- Engaged **2,000+** attendees through impactful hosting at major events, adeptly boosting the brand outreach and brand visibility.
- Contributed to the impactful marketing and social campaigns, adeptly increasing audience by nearly **10%** across all the platforms

ASSOCIATE MANAGER M&PR[6] (Nov' 23 - Mar' 24)

Co-managed and executed comprehensive marketing & public relations initiatives, enhancing brand reputation & visibility.

- Streamlined and optimized the critical operational workflows while actively fostering **team synergy** and improving team efficiency.
- Drove substantial participant engagement and maximized the outreach by applying targeted **marketing strategies**, in the events.
- Developed and maintained effective **databases**, adeptly ensuring **data integrity** accuracy, for efficient & strategic decision making.

SOCIAL IMPACT

IN-VERSE

- Managed In-verse, an IIT Madras E-Cell social campaign empowering artists to transform their creativity into sustainable careers.
- Organized an art expo connecting local artists with the public, showcasing their work and drawing a crowd of **2.5K - 3K** attendees.
- Led Alchemy podcast series with 5 established artists, reaching **10K+** Spotify listeners, and curated a social-betterment pitch event

UDAAN

- Played a key role in Udaan,an IIT Madras E-Cell initiative that reached **2L+** people through **20+** programs for rural entrepreneurship
- Helped uplift over **25 MSMEs**,**20 rural startups**, and **3 self-help groups** across Tamil Nadu as part of the Udaan campaign initiative.
- Orchestrated a Rural Expo featuring **20+** stalls, bringing together various **NGOs** and **SHGs** for promoting rural entrepreneurship.

EXTRA-CURRICULARS

Leadership & Service

- Led Nilgiri House as **House Captain** (8th Grade) to achieve 'overall champion' title, demonstrating **leadership** & **team building**
- Ensured the fair conduct of NDST/JDST examinations for 10th and 12th grade Navodaya scholars in 2024, serving as an invigilator.

Sports

- Represented the school as a **Cluster[7] level chess player** (9th grade), adeptly showcasing **strategic thinking** and **analytical skills**.
- Secured the Inter-House **Chess Champion title** (8th grade), and became the **runner up** (7th grade), demonstrating **consistency.**

Miscellaneous

- Achieved an impressive **2nd** position at **Cluster Level Science Exhibition** (8th Grade), Successfully competing against the students up to the 12th grade,& subsequently advanced to the **Regional Level**, highlighting strong **scientific aptitude** & **presentation skills**

[*] Ongoing courses | **[1]** Central Board of Secondary Education | **[2]** Joint Entrance Exam | **[3]** Navodaya Dakshana Selection Test |**[4]** vertical under shaastra and CFI
[5] Entrepreneurship Cell IITM | **[6]** Marketing and Public Relations (Vertical under E-Cell)| **[7]** Group of schools